

Course Number	COE 328
Course Title	Digital Systems
Semester/Year	Semester 3, Year 2
Section Number	1

Lab 1 - Introduction to CAD Tools

Assignment Title	Lab 1
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Theory:

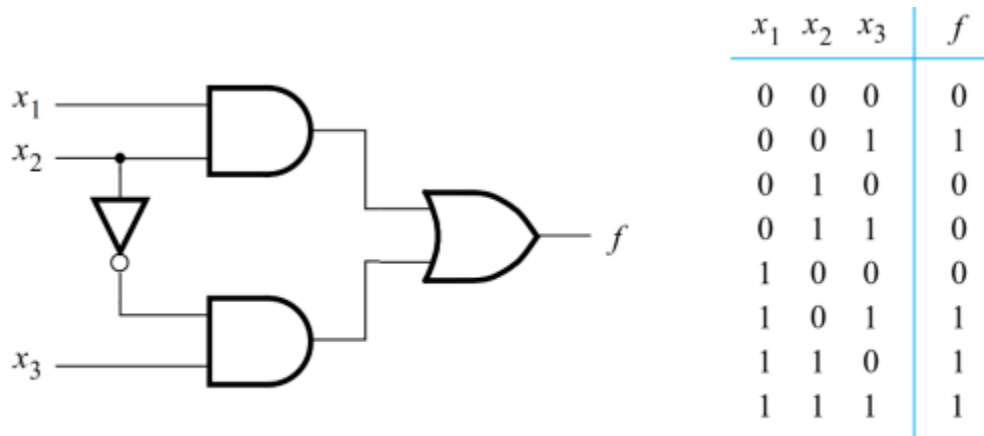


Figure 1

As we can see from the figure above, the given circuit and truth table are very simple and easy to understand. To build the given circuit we simply need 3 input pins (x_1 , x_2 , x_3), 2 AND gates, a NOT gate, and an OR gate. x_1 and x_2 would attach to one AND gate and x_3 to the other AND gate. For the missing part of the AND gate we would attach a NOT gate that would also attach to x_2 . Finally attach both AND gates to an OR gate. Finally, the schematic provides an output which is given by the pin named “ f ”. To understand the truth table we would need the function from the circuit which is $f = (x_1x_2) + (!x_2x_3)$. For (x_1x_2) the last 2 rows are the only ones with both $x_1x_2 = 1$ so $f = 1$. For $(!x_2x_3)$ we need to find the rows which $x_2 = 0$, $x_3 = 1$, so the 2nd row and the 6th row, f would = 1.

Figure 1: Screenshot of circuit diagram

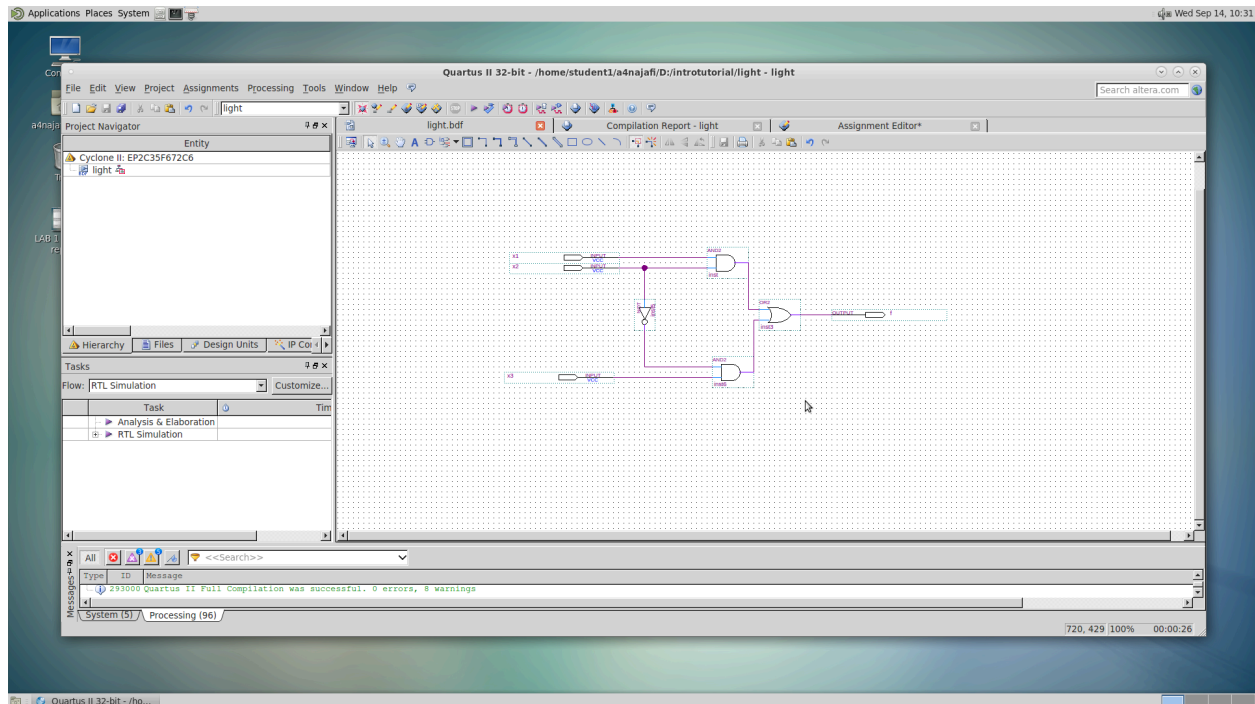


Figure 2: Screenshot of successful compilation

The screenshot shows the Quartus II 32-bit IDE interface with the Flow Summary window open. The Flow Summary window displays the results of the compilation process, including the Flow Status, Flow Settings, and Flow Summary. The Flow Status is 'Successful - Wed Sep 14 10:10:34 2022'. The Flow Summary table provides detailed information about the compilation process, including the number of logic elements, registers, pins, and memory bits used.

Flow Status	Flow Settings	Flow Summary
Successful - Wed Sep 14 10:10:34 2022	Quartus II 32-bit Version: 13.0.1 Build 232 06/12/2013 SP 1 5j Full Version	light
	Revision Name: light	Top-level Entity Name: light
	Family: Cyclone II	Device: EP2K35F672C6
	Timing Models: Final	Total logic elements: 1 / 33,216 (< 1 %)
		- Total combinational functions: 1 / 33,216 (< 1 %)
		- Dedicated logic registers: 0 / 33,216 (0 %)
		Total registers: 0
		Total pins: 4 / 475 (< 1 %)
		Total virtual pins: 0
		Total memory bits: 0 / 483,840 (0 %)
		Embedded Multiplier 9-bit elements: 0 / 70 (0 %)
		Total PLLs: 0 / 4 (0 %)

Figure 3: Simulation results circuit diagram

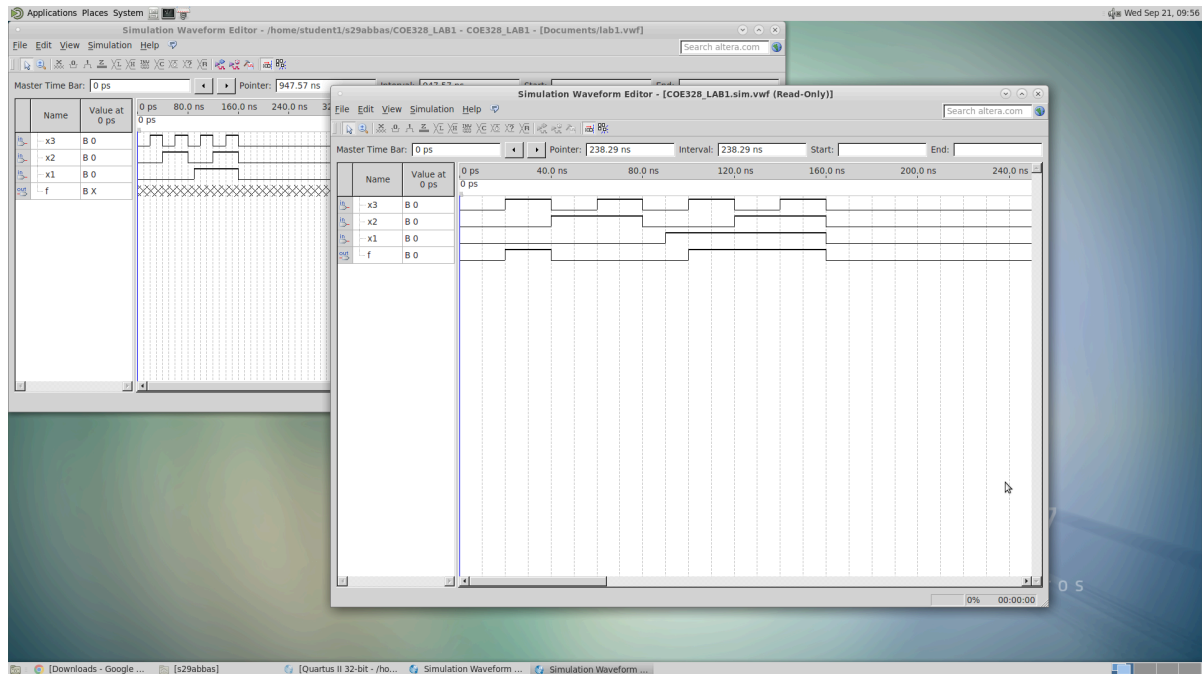


Figure 4: Screenshot of the first vhdl code

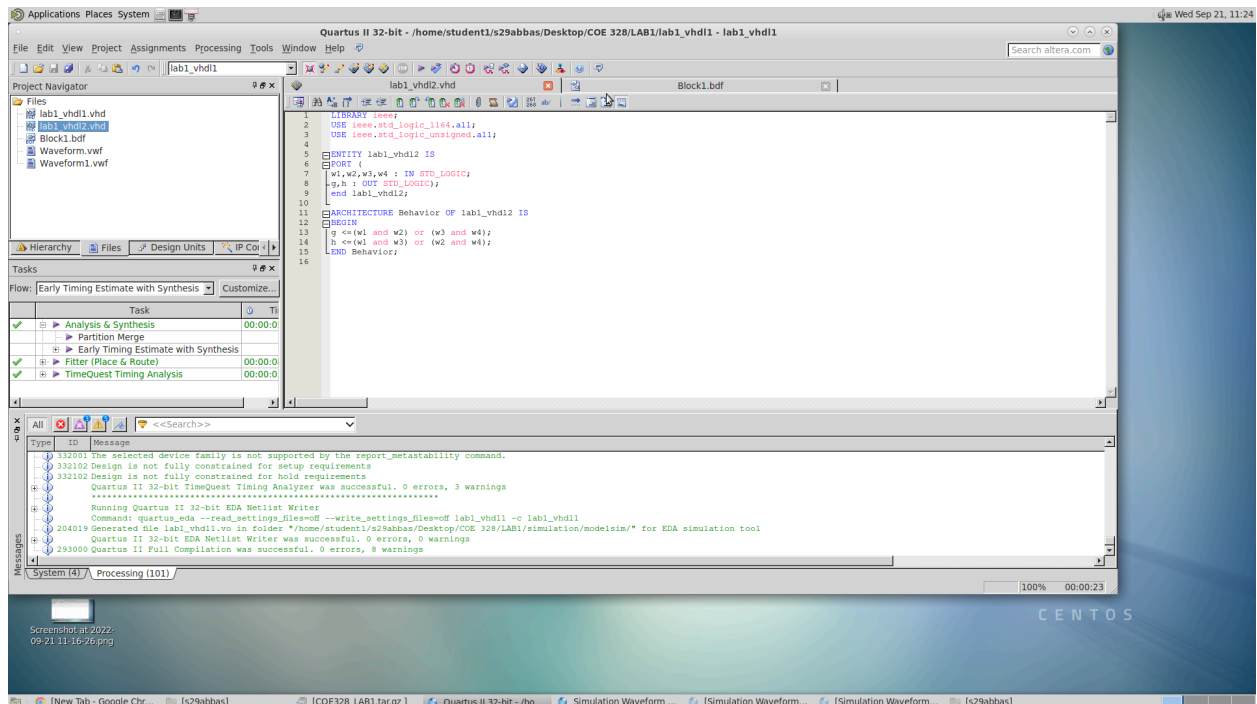


Figure 5: Screenshot of the second vhd code

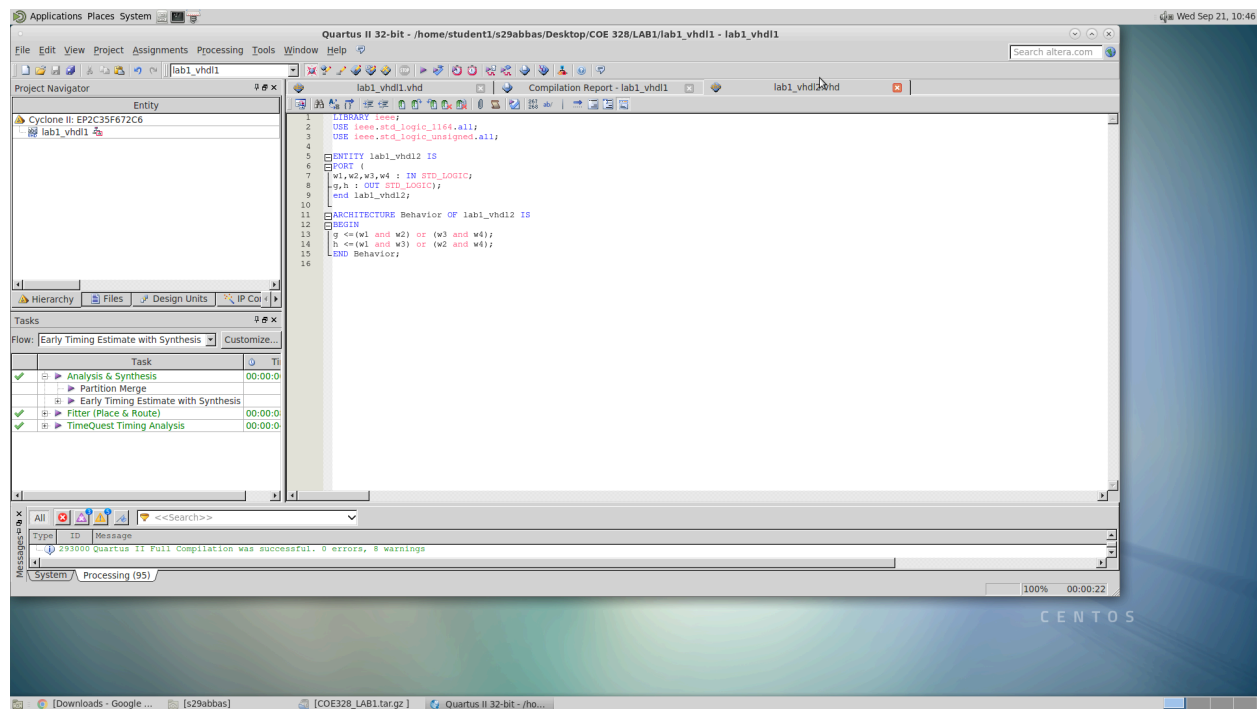


Figure 6: Screenshot of block diagram

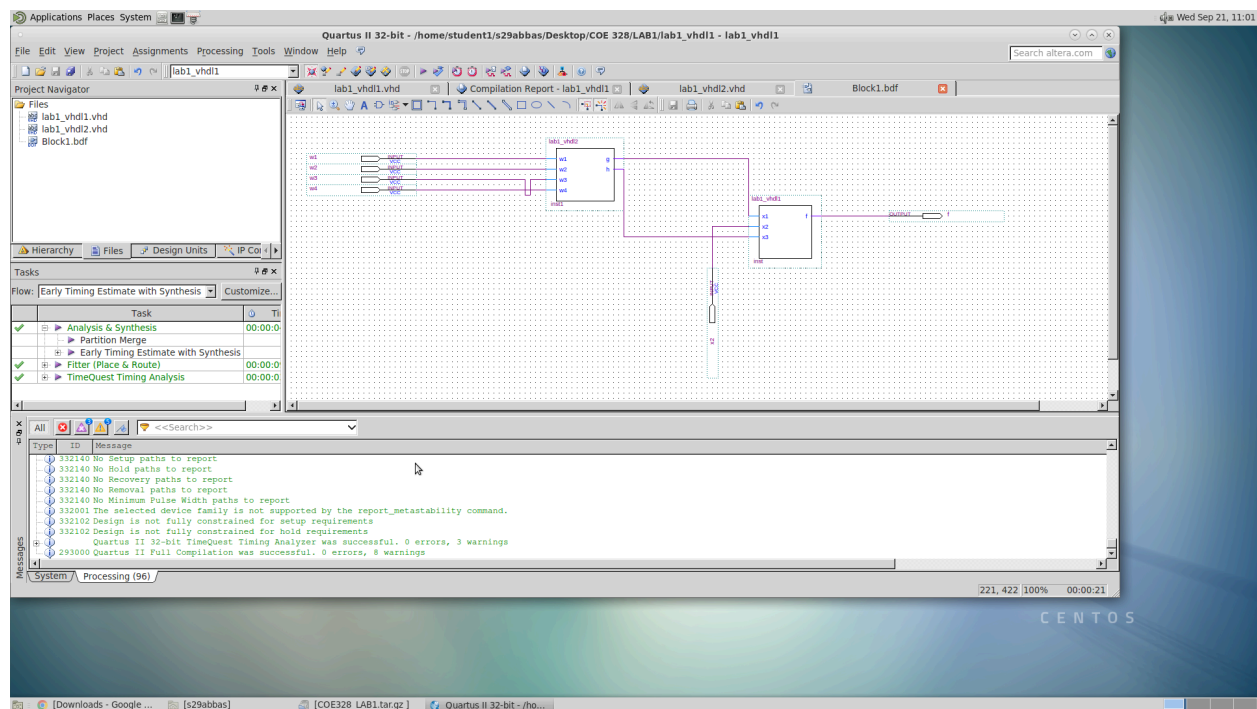
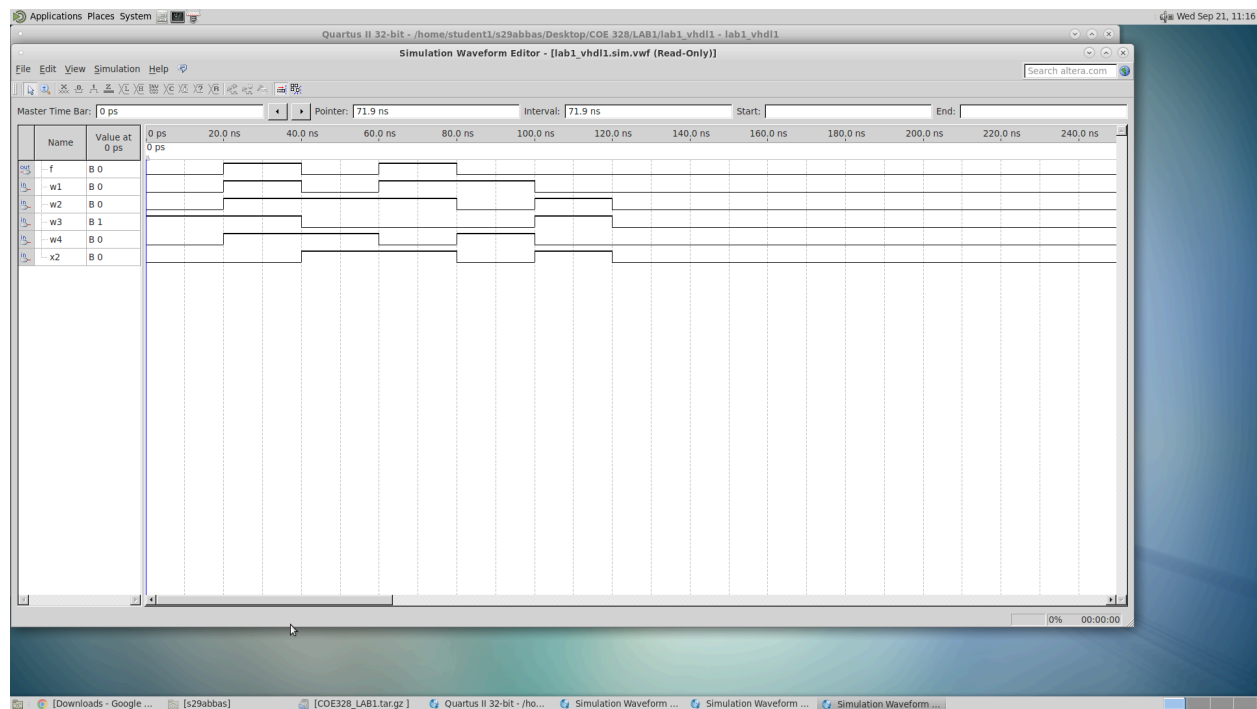


Figure 7: Simulation results of block diagram



Post-Lab:

The results of this simulation describe all the possibilities of this circuit. When the switch is turned on, the binary representation of this is expressed by “1”. Similarly when the switch is turned off this is represented by a “0”, thus the output function (f) shows the final position of the switch after going through the schematic. This can also be further analyzed using a truth table:

Table 1: Truth table of simulation

w1	w2	w3	w4	x2	f
0	0	1	0	0	0
1	1	1	1	0	1
0	1	0	1	1	0
1	1	0	0	1	1
1	0	0	1	0	0
0	1	1	0	1	0

Based on these results, the final position of the switch (f) can be predicted given that the input is arranged in a similar manner. However, this table does not provide all the possible outcomes, given that there are 5 input variables, there are a total of 32 possible combinations. ($2^n = 2^5 = 32$).

Conclusion:

In conclusion, this lab describes the process of visually crafting, analyzing, and interpreting a circuit diagram through the simulation/compilation of schematics and VHDL documents. This is applicable and a valuable skill to have for the purpose of testing and analyzing circuits without the need of any hardware materials. As an engineer, it is important to test and evaluate blueprints and schematics before starting a new project, therefore, the introduction to the quartus software proves to be beneficial in the future along with the introduction of logic gates and systems.